Appendix D: Uncertainty Quick Reference

GUM Method for Measurement Uncertainty

Step			Equations
1.		Determine the measurement model equation	$y = f(X_1, X_2 \ldots, X_n)$
2.		Determine mean $\overline{x_i}$ and uncertainty $u(x_i)$ for each input variable	
	a.	Find mean of n repeated measurements of variable x_i ($x_{i,\,k}$ is the kth measurement on variable X_i)	$\overline{x_i} = \frac{1}{n}\sum_{k=1}^{n} x_{i,k}$
	b.	Compute Type A standard uncertainty $u_A(x_i)$ for each variable	$\sigma_i{}^2 = \frac{1}{n-1}\sum_{k=1}^{n}(x_{i,k} - \overline{x_i})^2$ $u_A(x_i) = \frac{\sigma_i}{\sqrt{n}}$
	c.	Compute Type B standard uncertainty $u_B(x_i)$ given knowledge of measurement equipment. Common distributions and use cases:	
		Normal: (Calibration certificate provides uncertainty as $\pm U$ with k-value and/or confidence)	$u_B(x_i) = U/k$
		Uniform: (Equipment specification of ±U (without confidence given); Digital resolution of equipment, with least significant digit 2 U)	$u_B(x_i) = \frac{U}{\sqrt{3}}$
		Triangular: (Analog gauge resolution of equipment with smallest discernable interval U)	$u_B(x_i) = \frac{U}{\sqrt{6}}$
	d.	Combine Type A and Type B components for input variable x_i	$u(x_i) = \sqrt{u_A(x_i)^2 + u_B(x_i)^2}$
	e.	Determine degrees of freedom for variable x_i using Welch-Satterthwaite	$\nu_A = n - 1$ (Type A) $\nu_B = \infty$ (typical, Type B) $\nu_{\text{eff}} = \frac{u_c^4(y)}{\frac{u_A^4(y)}{\nu_A} + \frac{u_B^4(y)}{\nu_B}}$
3.		Compute combined standard uncertainty	
	a.	Compute sensitivity coefficients c_i, evaluate at the mean ($X_i = \overline{x_i}$)	$c_i = \left(\frac{\partial f}{\partial X_i}\right)$

(continued)

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Step			Equations
	b.	Compute standard uncertainty, $u_c(y)$, of y	$u_c(y) = \sqrt{\sum_{i=1}^{N} c_i^2 u^2(x_i)}$
4.		Compute expected value of measurement model	$\overline{y} = f(\overline{x}_1, \overline{x}_2, \ldots)$
5.		Compute effective degrees of freedom of y using Welch–Satterthwaite (with sensitivity coefficients)	$\nu_{\text{eff}} = \frac{u_c^4(y)}{\sum_{i=1}^{N} \left(\frac{(c_i u(x_i))^4}{\nu_i} \right)}$
6.		Determine coverage factor for confidence level p with ν_{eff} degrees of freedom (see *T*-table below)	$k = t_p(\nu_{\text{eff}})$
7.		Report result as value and expanded uncertainty. Typically use 2 significant digits of uncertainty and report expected value to same number of decimal places. Always provide *k*-value and confidence level	$U = \overline{y} \pm k\, u_c(y)$

Percentage Points of the *t* Distribution

Confidence level for two-sided confidence interval						
Degrees of freedom ν	68.27%	90%	95%	95.45%	99%	99.73%
1	1.84	6.31	12.71	13.97	63.66	238.80
2	1.32	2.92	4.30	4.53	9.92	19.21
3	1.20	2.35	3.18	3.31	5.84	9.22
4	1.14	2.13	2.78	2.87	4.60	6.62
5	1.11	2.02	2.57	2.65	4.03	5.51
6	1.09	1.94	2.45	2.52	3.71	4.90
7	1.08	1.89	2.36	2.43	3.50	4.53
8	1.07	1.86	2.31	2.37	3.36	4.28
9	1.06	1.83	2.26	2.32	3.25	4.09
10	1.05	1.81	2.23	2.28	3.17	3.96
11	1.04	1.80	2.20	2.25	3.11	3.85
12	1.04	1.78	2.18	2.23	3.05	3.76
13	1.04	1.77	2.16	2.21	3.01	3.69
14	1.03	1.76	2.14	2.20	2.98	3.64
15	1.03	1.75	2.13	2.18	2.95	3.59
16	1.03	1.75	2.12	2.17	2.92	3.54
17	1.03	1.74	2.11	2.16	2.90	3.51
18	1.03	1.73	2.10	2.15	2.88	3.48
19	1.03	1.73	2.09	2.14	2.86	3.45
20	1.03	1.72	2.09	2.13	2.85	3.42
25	1.02	1.71	2.06	2.11	2.79	3.33
30	1.01	1.70	2.04	2.09	2.75	3.27
35	1.01	1.70	2.03	2.07	2.72	3.23
40	1.01	1.68	2.02	2.06	2.70	3.20

(continued)

Confidence level for two-sided confidence interval						
Degrees of freedom ν	68.27%	90%	95%	95.45%	99%	99.73%
45	1.01	1.68	2.01	2.06	2.69	3.18
50	1.01	1.68	2.01	2.05	2.68	3.16
100	1.005	1.660	1.984	2.025	2.626	3.077
∞	1.000	1.645	1.960	2.000	2.576	3.000

Guardbanding

Symmetric Specification Limits

For specification limits $\pm$ SL and measurement uncertainty $\pm$ U, calculate TUR $= \frac{\text{SL}}{U}$. Ensure SL and U are both reported to the same confidence level. See Chap. 5 for discussion on when TUR-based guardbanding is appropriate.

TUR	Description	Guardband
>4	The measurement is adequate	N/A
1.5–4	Must be guard banded to reduce risk	$\text{SL}_{\text{GB}} = \text{SL}\sqrt{1 - \frac{1}{\text{TUR}^2}}$
<1.5	Measurement is not adequate	N/A

Asymmetric Specification Limits

In the case of asymmetric limits, given as Value + X/ $-$ Y, convert to absolute limits $\text{SL}_U = \text{Value} + X$ and $\text{SL}_L = \text{Value} - Y$, and the equivalent symmetric specification limits $\text{SL} = \frac{\text{SL}_U - \text{SL}_L}{2}$. Calculate TUR and guardband, if needed, as described above.

One-Sided Specification Limits

For a one-sided limit, the measurement must be >SL with no lower limit, or the measurement must be <SL with no upper limit. In this case, always guard band by the measurement uncertainty U.

$$\text{SL}_{\text{GB}} = \text{SL} + U \text{ (for a minimum limit requirement)}$$

$$\text{SL}_{\text{GB}} = \text{SL} - U \text{ (for a maximum limit requirement)}$$

Metrology Reference Table

This table provides sources for special topics that may be of interest to practitioners of metrology. Chapter numbers refer to chapters of this book.

Metrology topic	Sources
Accuracy of measurement methods	ISO 5725
Bayesian methods in metrology	Chapter 11, JCGM 108 (in progress)
Binary measurement systems	Chapter 11, AIAG measurement systems analysis reference manual
Conformity assessment	ISO/IEC 17065:2012
Curve fitting	Chapter 10, ASTM E3080-19, ISO/TS 28037:2010
Establishing traceability	Chapter 3, ISO/IEC 17025
Experimental design in metrology	Chapter 9, ISO/WD TS 23471 (in progress)
Generating random variates	Chapter 8, ISO 28640:2010
Interlaboratory comparisons	NCSLI RP-15, ISO/IEC 17043
Legal metrology	OIML V1 (VIML), OIML recommendations
Managing calibration laboratories	ISO 17025, ANSI Z540.3
Modelling and equation building	Chapters 9, 10, JCGM 103 (in progress)
Proficiency testing	ISO/IEC 17025, ISO/IEC 17043, ISO 13528
Reference materials	ISO guide 35, ISO guide 33
Risk in measurement decision making	Chapter 5, JCGM 106, NCSLI RP-18, ISO 10576-1
Setting optimal calibration intervals	Chapter 10, NCSLI RP-1, ILAC-G24/OIML D-10
The SI, reporting measurement units	Chapter 3, ISO 80000-1:2009, NIST SP-811
Software in metrology	NPL reports DEM-ES-010, DEM-ES-011, NCSLI RP-13
Verification of geometrical product specifications	ISO/TC 213, ASME Y14.43
Vocabulary of metrology	Chapter 2, ISO/IEC 99 (JCGM 200:2012, the VIM)